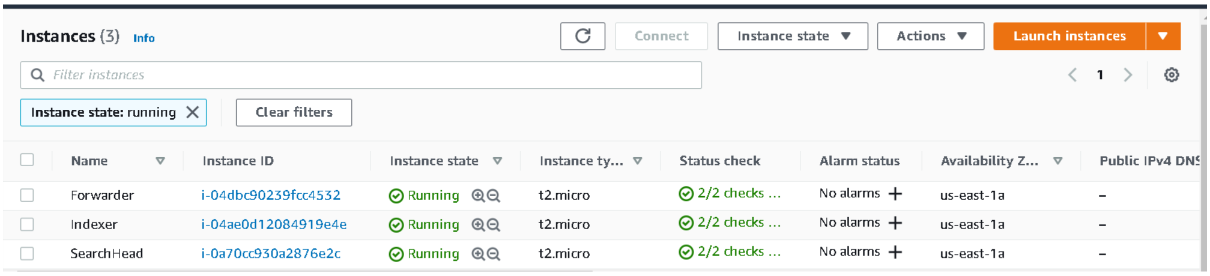
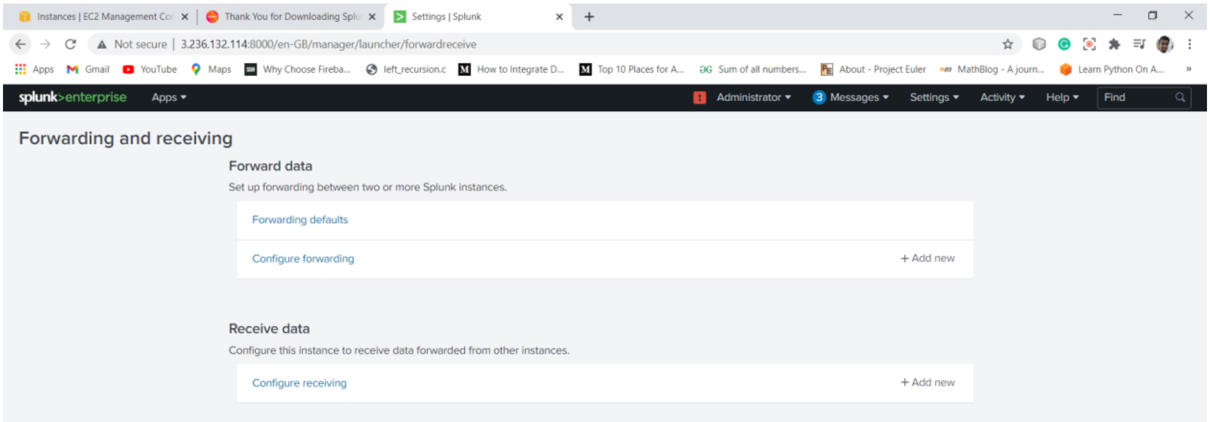
Distributed Architecture without Clustering:

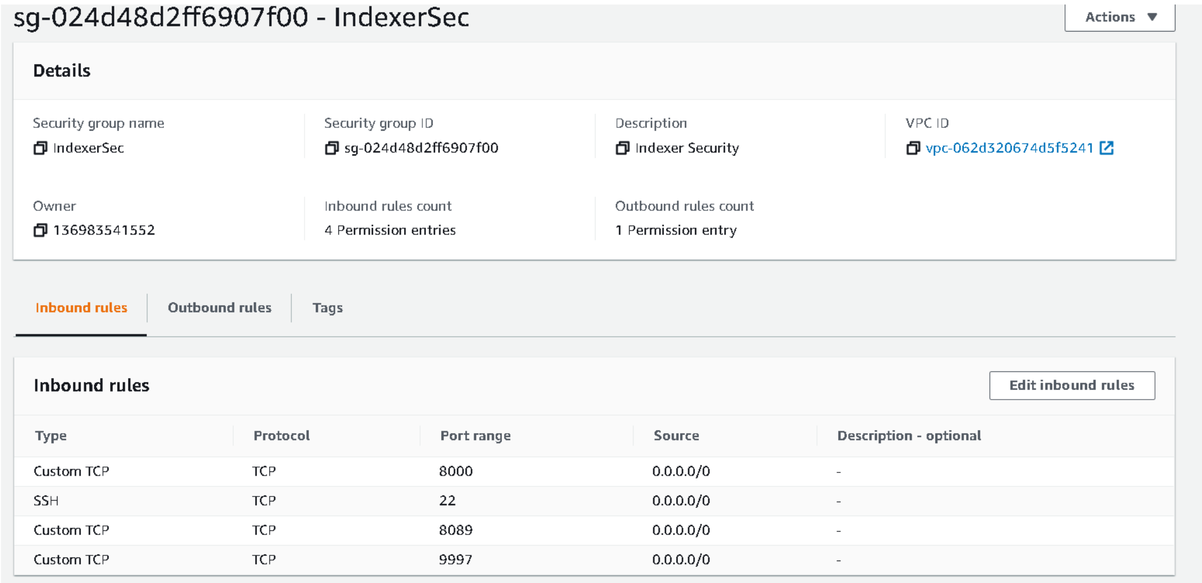
1)First create three instances one as Forwarder,second as Indexer and third as Search Head in public subnet for easiness to access . Install and configure Splunk Enterprise in instance you have selected as Search Head and Indexer.Install and configure Splunk Universal Forwarder in the instance you selected as Forwarder.

2) In inbound rules of security group of all the three instances allow traffic on port 8000 from source Internet(0.0.0.0/0) or you can be specific also.



3) SSH the Indexer open receiving port 9997 from Splunk web page going on *Forwarding and Receiving* and then open the port 9997 in the inbound rules for source Internet in security group of Indexer.





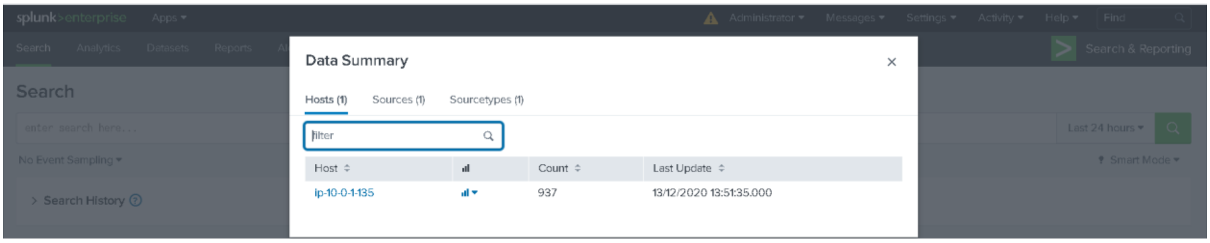
4)SSH into the forwarder and give command *./splunk add forward-server indexe_public_ip : port -auth username:password* .

```
[ec2-user@ip-10-0-1-135 bin]$ sudo ./splunk add forward-server 35.173.1.101:9997 -auth Forwarder:varun@123
Added forwarding to: 35.173.1.101:9997.
```

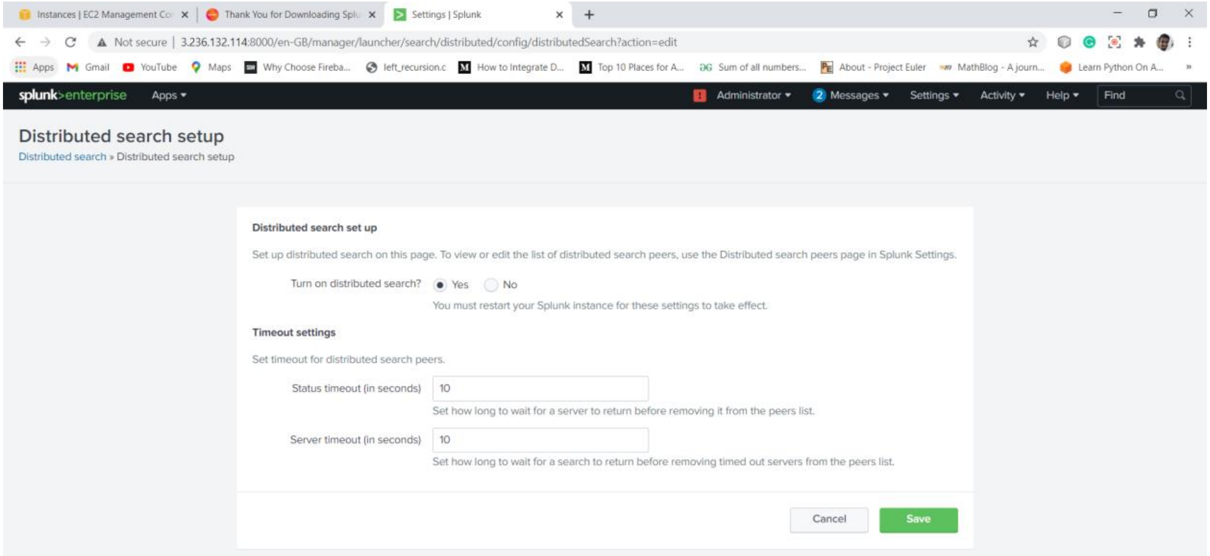
5) *./splunk add monitor path_of_logs_to_monitor*

```
[ec2-user@ip-10-0-1-135 bin]$ sudo ./splunk  add monitor /var/log/messages
Added monitor of '/var/log/messages'.
```

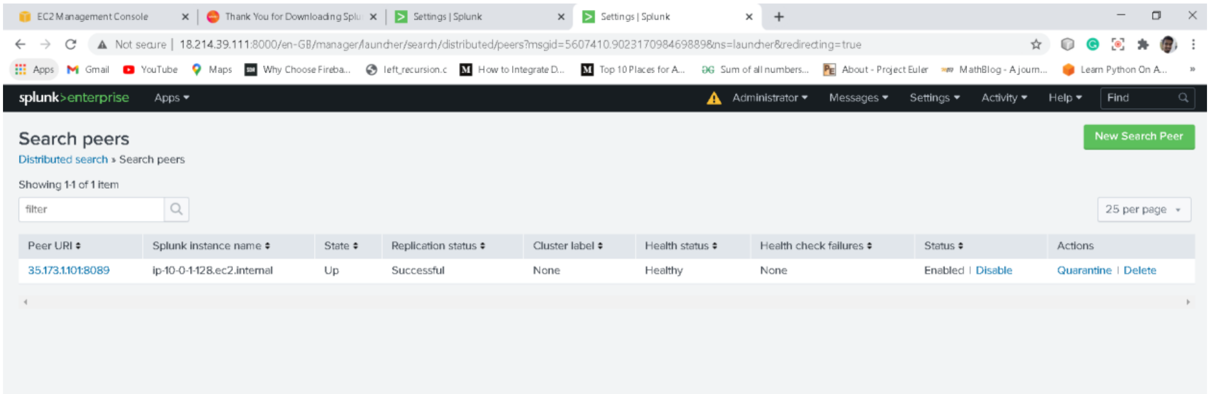
You can see that data is coming from forwarder to Indexer.



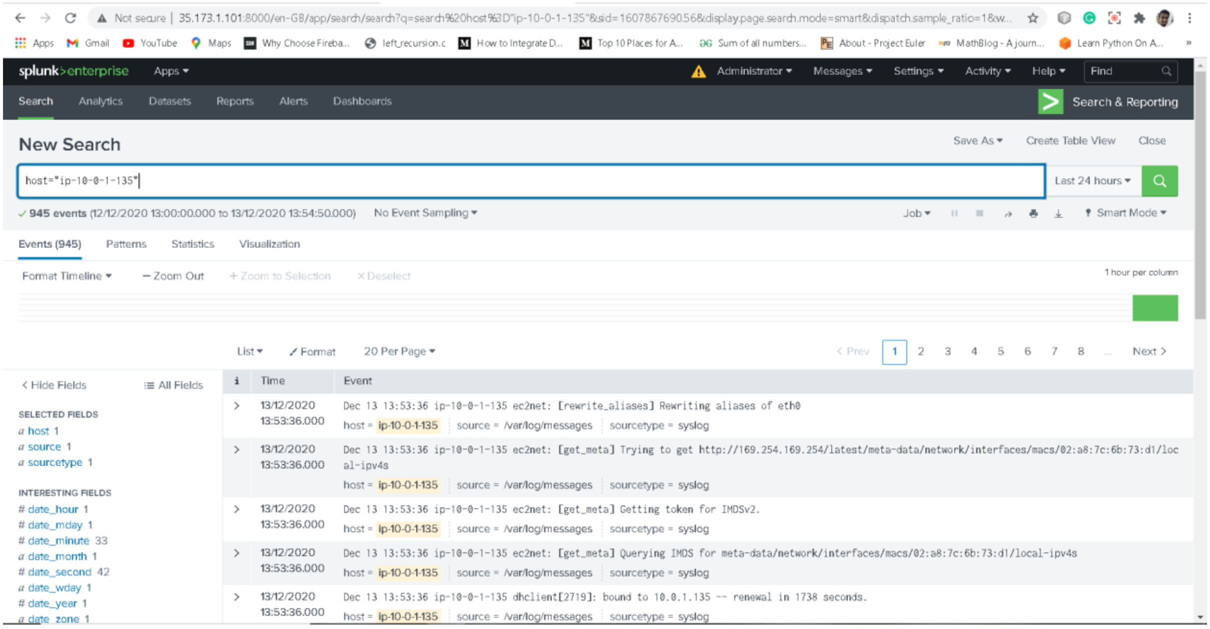
6)Check if the distributed search is on Search Head.Open the 8089 port of the indxer which will be used as Search peer.



7)Add it indexer as search peer.

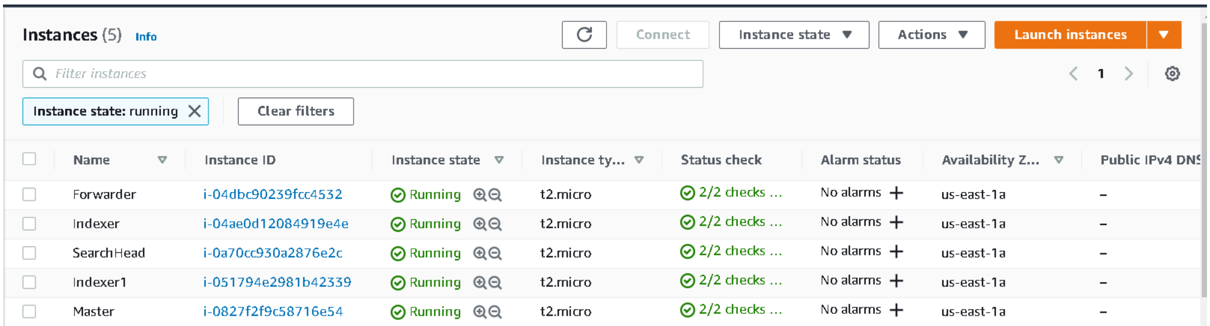


8) Now the forwarder data can be searched from Search Head instance.

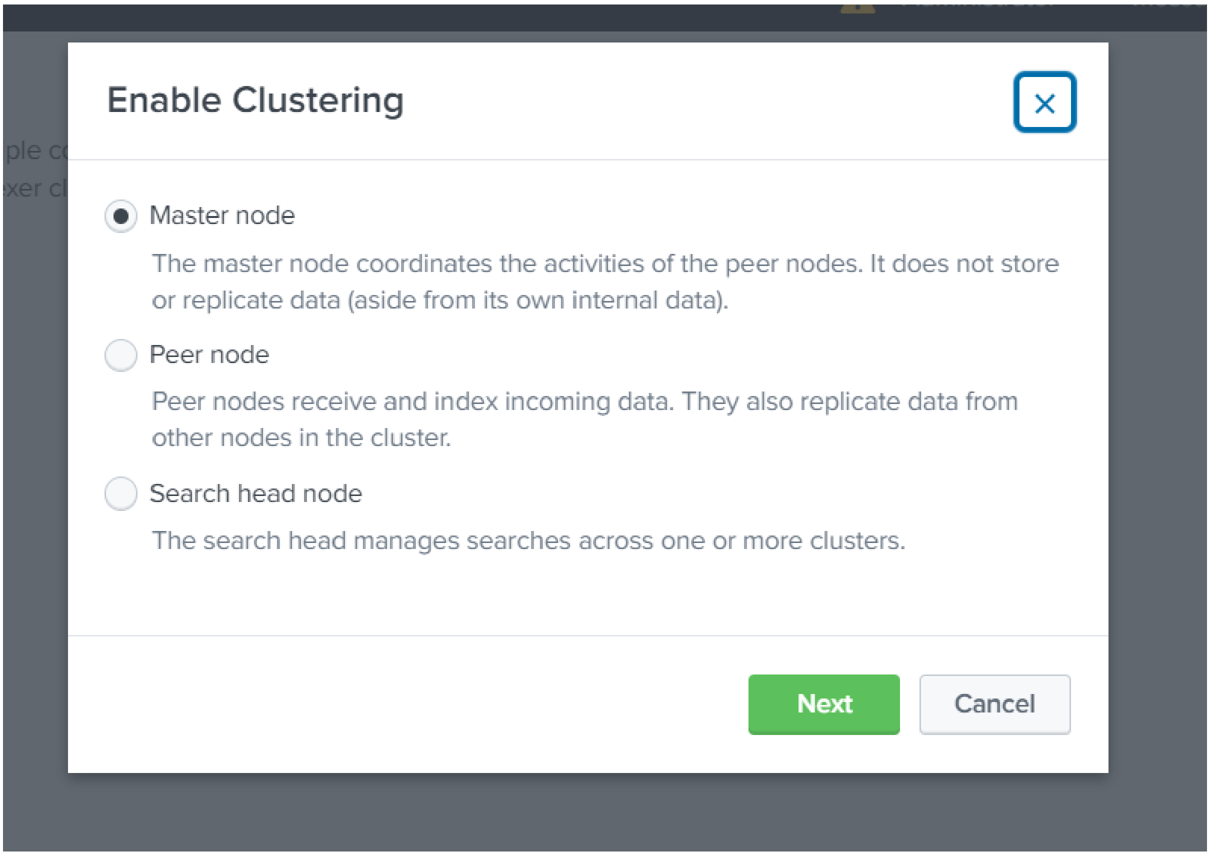


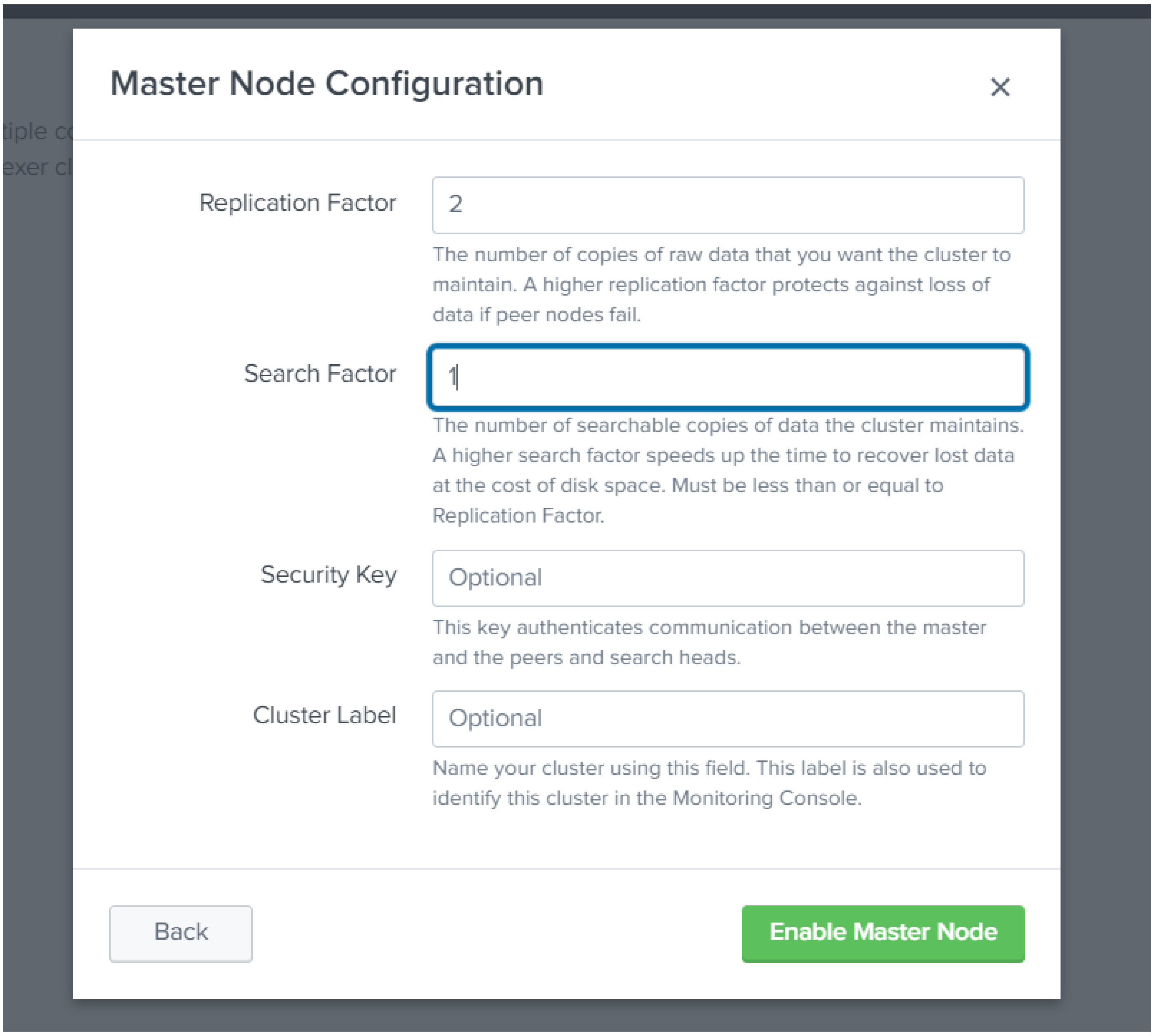
Index Clustering Architecture:

9) Create one more Indexer instance and a Master Node using security group used for previous Indexer.

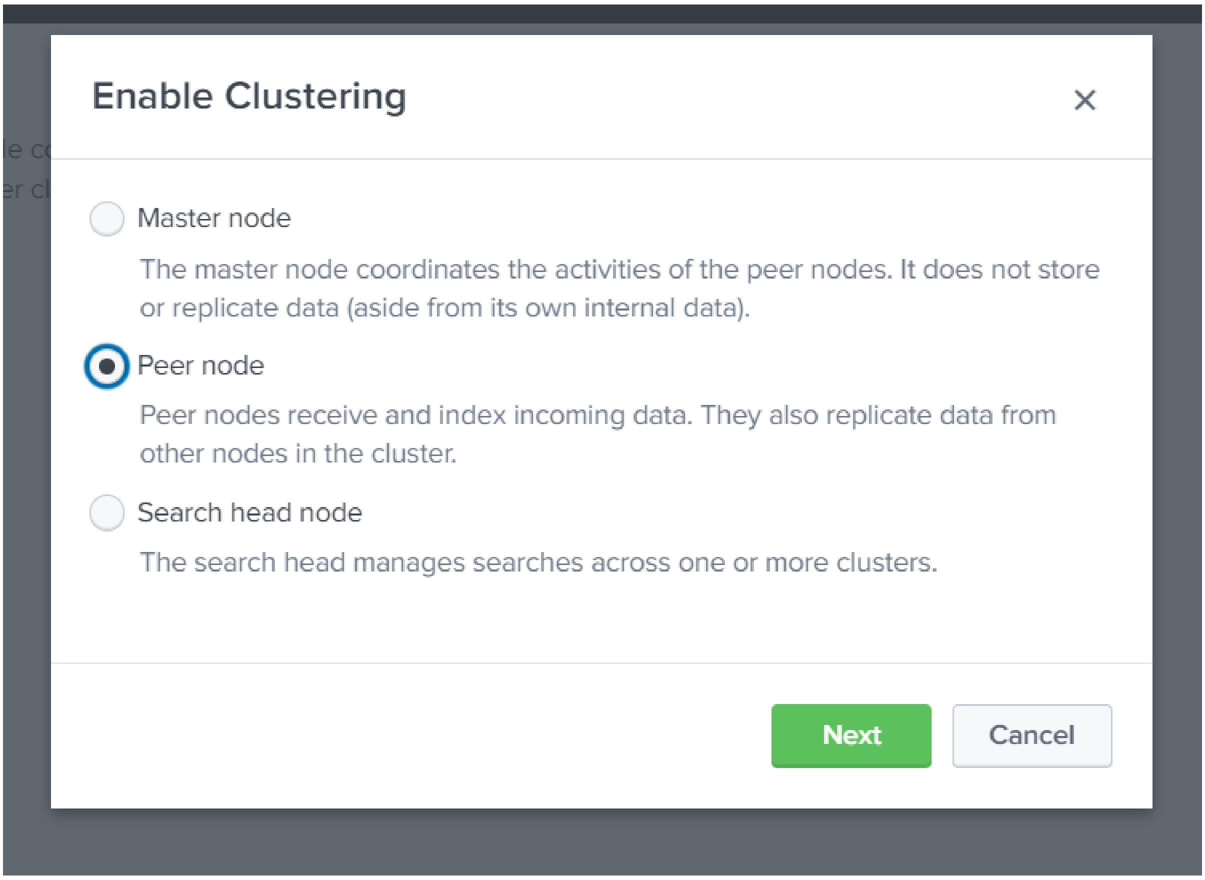


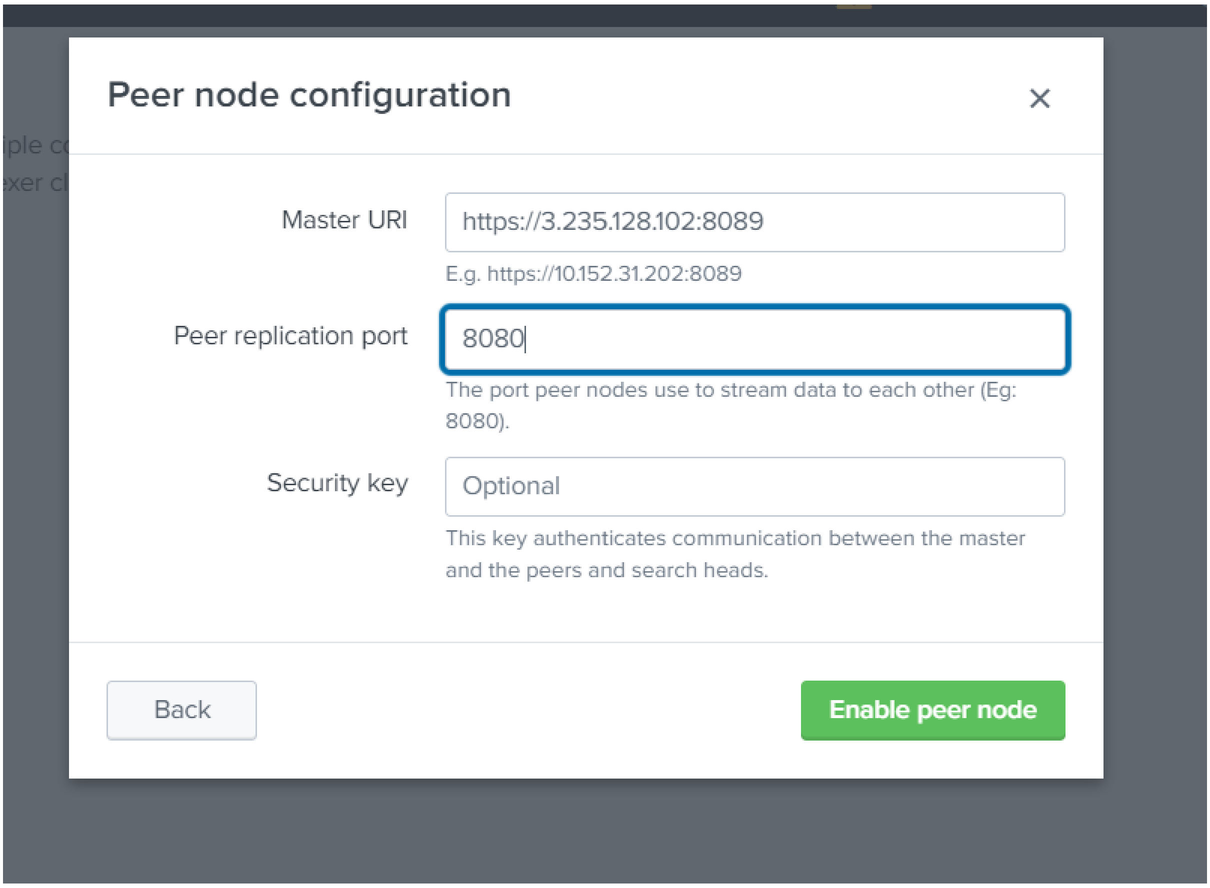
10) SSH Master node and start Splunk in its web UI go to index clustering in Settings and enable clustering. In configuration select the replication factor, search factor.


Enable Clustering
Master node
The master node coordinates the activities of the peer nodes. It does not store or replicate data (aside from its own internal data).
Peer node
Peer nodes receive and index incoming data. They also replicate data from other nodes in the cluster.
Search head node
The search head manages searches across one or more clusters.
Next
Cancel



11) SSH the all the Indexer which are to be made the part of the Index cluster. Enable the 8080 port in the Indexer security group which is going to be used as the replication port. Give the HTTPS URI of the Master to all the Indexers with their management port number.


Enable Clustering
Master node
The master node coordinates the activities of the peer nodes. It does not store or replicate data (aside from its own internal data).
Peer node
Peer nodes receive and index incoming data. They also replicate data from other nodes in the cluster.
Search head node
The search head manages searches across one or more clusters.
Next
Cancel


Peer node configuration
Master URI
https://3.235.128.102:8089
E.g. https://10.152.31.202:8089
Peer replication port
8080
The port peer nodes use to stream data to each other (Eg: 8080).
Security key
Optional
This key authenticates communication between the master and the peers and search heads.
Back
Enable peer node

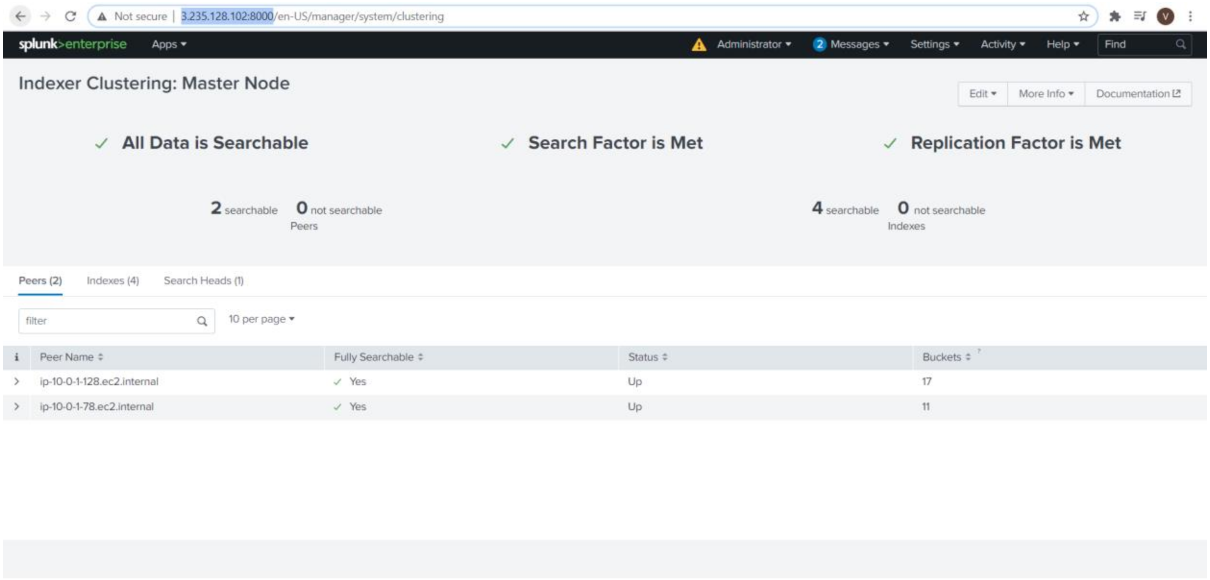
As soon as all the peers are configured the cluster is working

Clustering: Peer Node

Name ip-10-0-1-128.ec2.internal
Replication port 8080
Master location https://3.235.128.102:8089
Base Generation ID 1

Clustering: Peer Node

Name ip-10-0-1-78.ec2.internal
Replication port 8080
Master location https://3.235.128.102:8089
Base Generation ID 2



Search Head Clustering

12) Launch two more instance as Search head and a Deployer using security group used previously for search head. Open the port in inbound rules which is going to be used as replication port.

13) Go to server.conf of the Deployer using SSH client service and do shclustering configuration.

```
[ec2-user@ip-10-0-1-12 bin]$ cd ..
[ec2-user@ip-10-0-1-12 splunk]$ cd etc/system/local
[ec2-user@ip-10-0-1-12 local]$ vi server.conf
```

14) SSH each Search Head Instance go to bin of splunk and execute this command.

./splunk init shcluster-config -auth <username>:<password> -mgmt_uri <URI>:<management_port> -replication_port <replication_port> -replication_factor <n> -conf_deploy_fetch_url <URL>:<management_port> -secret <security_key>

```
ec2-user@ip-10-0-1-44:/opt/splunk/bin
[ec2-user@ip-10-0-1-44 bin]$ ./splunk init shcluster-config -auth SearchHead:varun@123 -mgmt_uri https://18.214.39.111:8089 -replication_port 9000 -replication_factor 3 -conf_deploy_fetch_url https://3.239.22.130:8089 -secret varun123
```

15) Make any one Search Head Captain.Using this command

./splunk bootstrap shcluster-captain -servers_list "<URI>:<management_port>,<URI>:<management_port>,..." -auth <username>:<password>

16) The cluster is up. See the status of any instance and the members using specific commands.

--see the cluster status

./splunk show shcluster-status -auth <username>:<password>

```
[root@ip-10-0-1-33 bin]# ./splunk show shcluster-status -auth searchhead3:varun@123

 Captain:
                              dynamic_captain : 1
                              elected_captain : Sun Dec 13 17:22:29 2020
                                           id : C1867A0E-0102-4C26-B6DB-37595B840997
                             initialized_flag : 1
                                        label : ip-10-0-1-33.ec2.internal
                                     mgmt_uri : https://34.235.150.67:8089
                        min_peers_joined_flag : 1
                         rolling_restart_flag : 0
                           service_ready_flag : 1

 Members:
        ip-10-0-1-172.ec2.internal
                                        label : ip-10-0-1-172.ec2.internal
                        last_conf_replication : Pending
                                     mgmt_uri : https://3.237.232.3:8089
                               mgmt_uri_alias : https://3.237.232.3:8089
                                       status : Up
        ip-10-0-1-33.ec2.internal
                                        label : ip-10-0-1-33.ec2.internal
                                     mgmt_uri : https://34.235.150.67:8089
                               mgmt_uri_alias : https://34.235.150.67:8089
                                       status : Up
        ip-10-0-1-44.ec2.internal
                                        label : ip-10-0-1-44.ec2.internal
                        last_conf_replication : Pending
                                     mgmt_uri : https://18.214.39.111:8089
                               mgmt_uri_alias : https://18.214.39.111:8089
                                       status : Up
[root@ip-10-0-1-33 bin]#
```

--see the member status

./splunk list shcluster-config -auth <username>:<password>

```
[root@ip-10-0-1-33 bin]# ./splunk list shcluster-config -auth searchhead3:varun@123
        config
                adhoc_searchhead:0
                async_replicate_on_proxy:1
                captain_is_adhoc_searchhead:0
                conf_deploy_fetch_url:https://3.239.22.130:8089
                cxn_timeout:60
                decommission_search_jobs_wait_secs:180
                disabled:0
                dispatching_mode:push
                dynamic_captain:1
                heartbeat_period:5
                heartbeat_timeout:60
                id:C1867A0E-0102-4C26-B6DB-37595B840997
                manual_detention:off
                max_peer_rep_load:5
                mode:dynamic_captain
                percent_peers_to_restart:10
                ping_flag:1
                preferred_captain:1
                quiet_period:60
                rcv_timeout:60
                rep_cxn_timeout:60
                rep_max_rcv_timeout:600
                rep_max_send_timeout:600
                rep_rcv_timeout:60
                rep_send_timeout:60
                replication_factor:3
                replication_port:9000
                replication_use_ssl:0
                restart_timeout:600
                rolling_restart:restart
                secret:********
                send_timeout:60
```